

Pranotosh Mandal

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PROFESSIONAL SUMMARY

M.Tech graduate in Data Analytics from IIT (ISM) Dhanbad with a strong foundation in Machine Learning, Deep Learning, and Generative AI. Skilled in designing scalable AI systems, developing LLM-powered applications, and deploying machine learning solutions using Python, SQL, FastAPI, PyTorch, Docker, AZURE and AWS.

TECHNICAL SKILLS

Programming	Python, SQL
GenAI & Agentic AI	Variational Autoencoders(VAEs), Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), AI Agents, Generative Adversarial Networks (GANs), Vision Transformers (VITs), Transformers, Prompt Engineering, Fine-tuning, LangChain, CrewAI
Domains	Machine Learning, Deep Learning, Computer Vision, Natural Language Processing (NLP), Transfer Learning, Generative AI
Frameworks & APIs	Scikit-Learn, PyTorch, FastAPI, Streamlit, N8N, REST APIs
MLOps & Cloud	Docker, Docker Compose, GitHub Actions (CI/CD), AWS (EC2, S3, IAM), AZURE, Linux, NVIDIA DGX Server, Render, Model Deployment, Huggingface
Databases & Vector Stores	MySQL, PostgreSQL, FAISS, Pinecone
Developer Tools	Git, GitHub, Jupyter Notebook, VS Code, Antigravity, Claude, Cursor, Power BI

PROJECTS

Text-to-SQL Agent | [↗ LIVE](#)

Python · LangChain · Groq (openai/gpt-oss-120b) · FastAPI · PostgreSQL · Streamlit · Docker · GitHub Actions · Render

- ▶ Built a production-ready AI agent using the LangChain ReAct framework that converts natural language questions into validated PostgreSQL queries; agent dynamically introspects the live database schema from `information_schema` at query time — enabling zero-code adaptation to any PostgreSQL database.
- ▶ Implemented a three-tool pipeline: (1) schema introspection, (2) SQL generation via Groq openai/gpt-oss-120b, and (3) SQL execution with a safety layer that blocks destructive statements (`DROP`, `DELETE`, `TRUNCATE`) — only `SELECT` queries are permitted.
- ▶ Served via FastAPI REST API and an interactive Streamlit dashboard featuring editable SQL, auto-generated charts (Bar/Line/Scatter), and CSV/SQL export; containerised with Docker Compose and fully automated CI/CD via GitHub Actions (pytest → Docker build → Render deploy).

AI Agent Chatbot | [↗ LIVE](#)

Python · LangGraph · LangChain · Groq · Gemini · Tavily · FastAPI · Streamlit · Render · GitHub Actions

- ▶ Developed a multi-model conversational AI chatbot enabling users to switch between Groq and Gemini LLMs at runtime; integrated Tavily API for optional real-time web search, grounding responses in current information beyond model training data.
- ▶ Architected stateful multi-step reasoning using LangGraph with persistent conversation memory; served via FastAPI and Streamlit UI, deployed on Render with a fully automated GitHub Actions CI/CD pipeline (test → build → deploy on every commit to main).

Multi-Source RAG Application | [↗ LIVE](#)

Python · LangChain · HuggingFace · Groq · Docker · FAISS · Streamlit

- ▶ Designed an end-to-end Retrieval-Augmented Generation (RAG) system supporting heterogeneous knowledge sources — PDF, plain text, web URLs, and YouTube transcripts — in a unified pipeline.
- ▶ Built document ingestion, embedding generation, and FAISS-based semantic retrieval pipelines; deployed through dockerization as an interactive Streamlit app for question answering and summarisation on Huggingface.

Fashion Recommender System | ↗ LIVE

Python · ResNet50 · FAISS · Streamlit · Computer Vision

- Extracted deep visual embeddings from a fashion image dataset using a pretrained ResNet50 backbone; indexed embeddings with FAISS for sub-millisecond nearest-neighbour similarity search.
- Built a Streamlit interface enabling users to upload a garment image and receive visually similar product recommendations, combining computer vision with efficient vector search.

House Price Prediction | ↗ LIVE

Python, XGBoost, FastAPI, Docker, GitHub Actions

- Built a production-ready regression API using XGBoost on the Ames Housing dataset; applied systematic feature engineering and hyperparameter tuning to maximise predictive accuracy.
- Containerised with Docker and served via FastAPI; implemented a GitHub Actions CI/CD pipeline to automate testing and ensure reproducible builds on every commit.

EDUCATION

M.Tech in Data Analytics

2024 – 2026

GPA: 7.82 / 10.00

IIT (ISM) Dhanbad · Dhanbad, Jharkhand

ACHIEVEMENTS & CERTIFICATIONS

- **Qualified GATE-2026** (Data Science and Artificial Intelligence) and **GATE-2024** (Mathematics)
- Neural Networks and Deep Learning — DeepLearning.AI Dec 2024
- Data Analytics Mentorship Program — CampusX Dec 2025
- Docker for Machine Learning — CampusX Feb 2026
- FastAPI for Machine Learning — CampusX Mar 2026

ADDITIONAL

Languages Bengali (Native) · Hindi (Fluent) · English (Professional)

Interests Badminton · Chess · Cricket · Football · Athletics